

Save New Examples for Testing

Project: Helpful Classroom Sorter

1. Save New Examples for Testing

Teach the learning cycle

Primary target: Separate examples used to choose a model from examples reserved for its final evaluation

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Build a Human Classifier

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated input format, selected rule and recorded output

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive. For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail. The contract defines accepted inputs; it does not list only the fitting records or establish prediction correctness.

Debug the Cause

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Compare expected and observed traces to locate the first difference

Reference example, not the only acceptable wording: Expected trace: 3, 5, 10, 9. First difference is the doubling checkpoint: 10 expected versus 8 observed. The later 8 minus 1 equals 7, so the observed subtraction is consistent with its input. Investigate the earlier step before claiming another fault.

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an ordered algorithm using the current value after every instruction

Reference example, not the only acceptable wording: The trace is 3, 7, 14, 9. The doubling step uses 7, the current value after adding 4 to 3. It must not double the initial 3. Accept a table, arrows or clearly labelled sentences that preserve the order.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Concept to explain

A fair test uses examples not used to create the decision process.

Term · Meaning

Term: training set

Meaning: examples used to create or adjust a process

Term: test set

Meaning: untouched examples used for final checking

Term: holdout

Meaning: data deliberately kept separate

Term: data leakage

Meaning: test information influences design

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Make the order of evidence visible. The lesson table is a retrospective worked example, not an unseen student test. W4 is a separate staged exercise: do not reveal its labels before predictions are committed. Avoid calling ordinary input validation a validation set here; the focus is development versus a held-out final check.

Readiness answer

Compare the prediction with an appropriate recorded label. If the label helps choose the threshold, that example has participated in fitting or model selection; it is not independent evaluation evidence for the resulting model.

Development and reserved tower cards from the worked example, marked as a completed experiment

RibbonA input cards 5, 7, 9, 11 cm and a covered label sheet: short, short, long, long

Paper for a dated frozen-rule record and separate predictions, labels and outcome columns

End goal for the pictorial case

By the end, I can freeze a model before labels are revealed and report a bounded test result.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT all labeled cards

PROCESS separate training and untouched testing before rule creation

OUTPUT honest performance evidence on new cards

CORE IDEA A fair test uses examples not used to create the decision process.

UNDER THE HOOD Evaluation on training data is optimistically biased. A holdout test set estimates behavior on unseen examples only if it remains untouched during design. Repeatedly tuning to test results leaks information and turns the test into training.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

STEP 1 — OBSERVE all labeled cards

STEP 2 — TRACE separate training and untouched testing before rule creation

STEP 3 — CONCLUDE honest performance evidence on new cards

MODEL REASONING Randomly or deliberately reserve representative cards first. Create the decision process using only the learning pile. Open the test pile once for evidence, record every result, and do not call the revised process tested until a new holdout exists.

DO NOT OVERCLAIM A test set stops being independent when its answers guide revisions.

What new evidence could change this answer?

Concrete teaching case

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

Exact explanation to model

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen

performance. Keep final test answers separate.

Explain the added worked example

A rule chosen to fit examples may work differently on new cases. To investigate that difference, reserve a set of examples before development. Use the development examples to choose the model. Keep the final test examples separate until the model is fixed, then evaluate that recorded model.

In this paper activity, a teacher keeps the reserved examples in an envelope. Record the input contract, feature, category rule and model version before opening it. Once the input values are revealed, predict using the frozen model before revealing their comparison labels. This prevents copying the labels into the predictions.

The worked example below describes a completed experiment, so its test labels are visible for discussion. It is not an unseen test for you. In the workbook experiment, the teacher will supply a separate set of comparison labels after you have saved your predictions. All data are imaginary classroom records.

In the completed experiment, development towers at heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Candidates predict tall at or above 5 or 6 cm. Both fit all four labels; the stated smaller-threshold tie rule selects 5. The team freezes model TowerA with threshold 5.

The reserved inputs are different records at heights 3, 5, 7 and 9 cm. TowerA accepts whole cm heights from 1 through 10. It predicts short, tall, tall and tall. These predictions are saved before the reserved labels are opened. The comparison below records what happened, including the failure.

Key terms

Term · Meaning

Term: Development examples

Meaning: Examples available for fitting, comparing or revising a model

Term: Held-out test set

Meaning: Examples reserved from development for evaluating a fixed model

Term: Freeze

Meaning: Record a model and stop changing it for a particular evaluation

Term: Test leakage

Meaning: Information from a reserved test influencing model development

Term: Evaluation record

Meaning: A record of the tested model, predictions, comparison labels and results

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

3 cm · Short · Short

3 cm: 5 cm

Short: Tall

Short: Short

3 cm: 7 cm

Short: Tall

Short: Tall

3 cm: 9 cm

Short: Tall

Short: Tall

TowerA agrees on three of these four reserved examples and disagrees at 5 cm. That is evidence about this fixed model on this small set. It does not prove a success rate for every future tower. The labels, range of examples and way records were selected also matter.

After seeing the 5 cm label, the team changes the threshold to 6 and names the revision TowerB. TowerB agrees on all four of these now-seen examples. The change may be useful, but those examples have influenced its development. Four agreements on them are not an independent final test result for TowerB. Keep TowerA's original three-of-four record.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Save New Examples for Testing

A fictional envelope exercise evaluates a fixed pencil-length sorter.

The facts we will use

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after predictions, are short,long,long,long.

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Save New Examples for Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze

Write PencilP and its boundary before opening the label envelope.



2. Predict

For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references

Envelope labels are short,long,long,long.



4. Compare

Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING

After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Freeze

What happens and why: Write PencilP and its boundary before opening the label envelope.

Step: 2. Predict

What happens and why: For 3,5,6,9 cm predict short,short,long,long.

Step: 3. Reveal references

What happens and why: Envelope labels are short,long,long,long.

Step: 4. Compare

What happens and why: Three of four agree; the disagreement is at 5 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The prediction record must exist before the labels are seen. PencilP scores $3/4=75\%$ agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

What this example does not establish

Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Change one thing in the pictured case

Change: After feedback, change the boundary to 5.

Prediction: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is evidence of repair rather than unseen performance.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Cards A–F are used for practice. G and H remain sealed.

Write the question to explain: Which can provide a fresh test, and under what condition?

Reveal after predictions: G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Keep for an independent attempt: You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

TowerA agreed on three of four original reserved records, failing at 5 cm. TowerB agreed on all four after their labels informed its revision; this is performance on development evidence. Freeze TowerB before evaluating fresh appropriate reserved examples. Shuffling cannot undo prior use. Exit: the team selected using test scores, so it measured a development comparison and still lacks an independent final check of the winner.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve the difference between accepted input and checked prediction. Ask when a label becomes part of development.

Stage: 7 to 20 minutes

Teacher action: Trace the three evidence stages in the diagram. Explicitly identify the lesson table as an already completed experiment.

Stage: 20 to 33 minutes

Teacher action: Compare TowerA predictions and revealed labels. Complete W1 and W2, preserving the original failure when TowerB is introduced.

Stage: 33 to 45 minutes

Teacher action: Use W3 to show how a score can influence model selection without exposing individual labels. Discuss why shuffling is not a repair.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve the frozen-model and saved-prediction requirements. Keep the RibbonA label sheet covered.

Stage: 8 to 23 minutes

Teacher action: Complete the guided task and plan the two statements for TowerA and TowerB. Describe what fresh evidence would be needed.

Stage: 23 to 38 minutes

Teacher action: Collect W4 predictions independently before revealing short, short, long, long. Students append results without erasing predictions, then explain the revised threshold.

Stage: 38 to 45 minutes

Teacher action: Use W5 and the exit to distinguish independence from coverage. Keep original and revised result claims side by side.

Reduce help deliberately

Completion task: TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its predictions, agreements and disagreement. Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

Teacher check: TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection. Report its four agreements as a result on examples used in development, preserving TowerA's original test record.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Keeping final tests unseen. Proposed explanation: "A new file makes old records unseen". Decide whether this explanation is supported. Point to the step or supplied fact that decides;

explain your repair if needed.

Expected correction: Trace whether records or copies already influenced development, regardless of their file names.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does 4/4 on the replay replace the original 3/4 result? Explain.

Reasoned teacher answer

No. Keep 3/4 as the frozen version's result; record 4/4 separately as a revised-version replay on known cases.

Changed-case reasoning: The revised rule matches all four known labels but needs a different final test. These labels influenced the revision, so replaying them is evidence of repair rather than unseen performance.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Split card data into learn and test piles before sorting.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR A card remains unseen after it has been used for teaching.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A new file makes old records unseen: Trace whether records or copies already influenced development, regardless of their file names.

Only labels copied into training can leak: A test score can guide rule selection even when individual labels remain hidden.

Testing means the model can never improve: Improve openly, preserve the original result, and obtain fresh independent evidence for the revision.

A perfect reserved score proves universal correctness: Check sample size, variety, relation to development records and intended use; keep the conclusion within the evidence.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Independent question

You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

Expected independent answer

D and E provide new evidence. If D is shown during teaching, it is no longer an unseen test example; keep E or obtain another fresh card for testing.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A card remains unseen after it has been used for teaching.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. They became development evidence when they influenced revision. Reserve new independently labelled cases for final evaluation. If none are available, disclose that limitation and report results as development results, not unseen performance.

Explanations for the additional workbook examples

W1 Assign the evidence its role

For each event, state whether examples are being used for development or independent evaluation of a fixed model: comparing candidates on fitting labels; scoring a frozen model on previously reserved examples once; changing a threshold after reading a test score. Explain your last choice.

Comparing candidates is development. Scoring a frozen model on previously reserved examples is evaluation, provided they did not influence its development. Changing a threshold after reading the test score uses that test information for development of the revision; the same set is no longer independent of that revision.

W2 Preserve the original result

TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its predictions, agreements and disagreement. Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection. Report its four agreements as a result on examples used in development, preserving TowerA’s original test record.

W3 Detect a less visible leak

A team leaves reserved labels out of its fitting table but tries six rules and reads a test score for each. It chooses the highest-scoring rule. Explain how information still reached model selection. Would renaming or reordering the same test records repair the problem?

The scores depend on the reserved labels and guide the choice among rules, so test information influences selection indirectly. Omitting the labels from a fitting table is insufficient. Renaming or reordering records does not remove that information; reserve new appropriate examples for an independent final check.

W4 Run a separate ribbon experiment

Before receiving any test labels, freeze RibbonA: whole cm lengths 1 through 12, long at least 6 cm and

short otherwise. Save predictions for 5, 7, 9 and 11 cm. Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement. If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

Saved predictions are short, long, long, long. Reveal labels only after predictions are saved: short, short, long, long. RibbonA has three agreements and one disagreement at 7 on four reserved records. Threshold 8 then gives short, short, long, long, but it was chosen after feedback. Its four agreements are on now-seen development examples, not a new independent test. Do not alter the saved RibbonA predictions.

W5 Audit a proposed final set

A class reserves ten copies of one easy development example in a new file. The model agrees with all ten labels. Explain two reasons this gives weak evidence about new cases, and propose a better final evaluation plan without promising perfect future performance.

Copies repeat a case already used in development and provide almost no variety; a new file does not make them new evidence. Reserve distinct relevant cases before model choices, keep related copies together, freeze the model, save predictions, reveal labels and report every outcome. Include cases relevant to intended use and state that a finite test cannot guarantee all future results.

Assess independent transfer

Selected case: Before receiving any test labels, freeze RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save predictions for 5, 7, 9 and 11 cm. Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement. If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

Selected case answer

Saved predictions are short, long, long, long. Reveal labels only after predictions are saved: short, short, long, long. RibbonA has three agreements and one disagreement at 7 on four reserved records. Threshold 8 then gives short, short, long, long, but it was chosen after feedback. Its four agreements are on now-seen development examples, not a new independent test. Do not alter the saved RibbonA predictions.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 16: Boundaries and 'Not Sure'. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use two envelopes and a timeline with a clear freeze point. Read the steps aloud, preserving the learner's original predictions. Extension: explain why selecting the best of many test scores may give a flattering result even without deliberate cheating.

W4 is secure with saved predictions, three agreements, the 7 cm failure, a limited claim and a correct account of post-test revision. If labels were seen early, record the exercise as coached practice. For an uncoached retry freeze BandA at long at least 4, use inputs 2, 4, 6, 8, then reveal short, short, long, long: three agreements, failure at 4.

Week 16 develops explicit uncertainty and review paths when available evidence does not justify a confident category or action.

Plan the delayed return

Suggested return points in this level: ai-1/16, ai-1/18, ai-1/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Maintain dated sorter versions and separate development examples from reserved project-evaluation cases.

7. Prepare, teach and assess

Teaching target: Separate examples used to choose a model from examples reserved for its final evaluation

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its predictions, agreements and disagreement. Then state

why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

Reasoned practice answer: TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection. Report its four agreements as a result on examples used in development, preserving TowerA's original test record.

Check every learner before discussion: Use the worked case for Keeping final tests unseen. Proposed explanation: "A new file makes old records unseen". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Trace whether records or copies already influenced development, regardless of their file names.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: After feedback, change the boundary to 5. Require a prediction and a reason before comparing. Expected result: The revised rule matches all four known labels but needs a different final test. Reason: These labels influenced the revision, so replaying them is evidence of repair rather than unseen performance.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.